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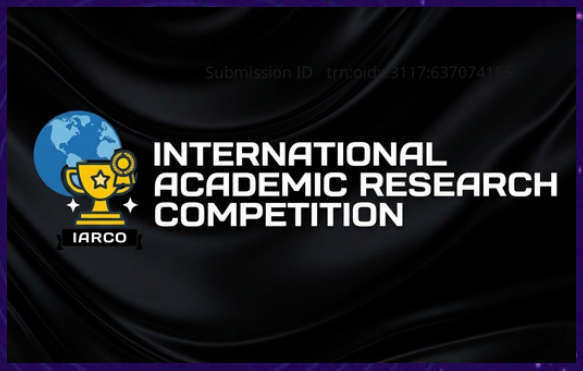
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TESTING CHROMATIN DISORGANIZATION AS A TOPOLOGICAL BIOMARKER FOR PEDIATRIC HIGH-GRADE GLIOMA USING REAL HI-C DATA

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Testing Chromatin Disorganization as a Topological Biomarker for Pediatric High-Grade Glioma Using Real Hi-C Data

Abstract

Diffuse midline glioma is one of the most lethal pediatric cancers, driven in most cases by a single point mutation in histone H3 (H3K27M) that disrupts chromatin structure rather than a single gene. No molecular biomarker derived from three-dimensional genome architecture exists for this tumor yet. In an earlier exploratory phase of this research, I applied H0 persistent homology, a technique from topological data analysis, to eighty simulated Hi-C chromatin structures, and cancer-associated chromatin came out far more heterogeneous than normal, including an unpredicted bimodal pattern where hyper-stable and fragmented domains coexisted in the same population. That finding relied entirely on simulated data, a limitation named directly by external review. This proposal is the confirmatory phase: it applies persistence-based chromatin analysis to real H3K27M-glioma tissue for the first time. It tests whether the same signature holds up in real patient Hi-C data, building on that earlier pipeline. It also replaces the original geometry-only reconstruction with a method that models physical chromatin connectivity, and applies bootstrap resampling to check how reliable each sample's result is.

Introduction

Diffuse midline glioma, which most commonly arises in the brainstem of children, carries a prognosis that hasn't meaningfully improved in decades. Median survival remains 8 to 12 months from diagnosis, with fewer than 10 percent of patients surviving two years [12]. No approved therapy targets the tumor's defining molecular lesion directly [1], and no biomarker derived from its three-dimensional structure exists either, leaving clinicians without a structural tool to guide risk stratification or trial eligibility while the disease's short timeline keeps advancing. That lesion is a mutation in histone H3 known as H3K27M, which inhibits the Polycomb repressive complex 2, disrupting a chromatin-modifying enzyme complex and changing how DNA is packaged and which genes are expressed [1], [10]. The mutation acts on chromatin architecture rather than a single gene, and its three-dimensional consequences may reveal structural features that gene expression data alone can't capture.

Hi-C sequencing measures how often pairs of genomic regions touch each other inside the folded nucleus, and this contact data can be converted into three-dimensional coordinates for structural analysis [2]. Persistent homology tracks how clusters of points merge as a distance threshold grows: each cluster has a birth, when it forms, and a death, when it merges into another, and the gap between the two, its persistence, measures how long a structure stays spatially distinct [5]. Applied to Hi-C-derived coordinates, this method can measure domain stability in ways single-scale contact-frequency statistics cannot.

In that earlier exploratory phase, evaluated by judges of another student research competition, I applied H0 persistent homology to eighty simulated Hi-C structures, forty normal and forty cancer, generated from empirically derived contact parameters. Cancer-associated structures showed a 44 percent increase in mean persistence and a 36 percent increase in persistence standard deviation, both significant. There was also an unpredicted bimodal pattern: hyper-stable domains coexisted with genome-wide fragmentation while total domain count stayed the same. Persistence-derived features made up eight of the fifteen most discriminative biomarkers in a random forest classifier. External review flagged three gaps: reliance on simulated data, a

reconstruction method that ignores basic polymer physics, and thin coverage of recent topological-genomics literature. This proposal answers all three, with real tissue, a connectivity-aware reconstruction, and the citations below. It is the second, confirmatory phase of a research arc focused on this specific cancer type, following the earlier simulated-data study.

Research Questions and Hypotheses

This study is built around one specific, falsifiable question: does the bimodal H0 persistence heterogeneity signature found in simulated cancer chromatin also appear in real Hi-C data from H3K27M-driven pediatric high-grade glioma, compared against real normal brain tissue? Persistent homology produces one concrete number, the standard deviation of persistence, and in real data that number either separates the two conditions or it doesn't, a result that would move this signature toward a usable structural biomarker, though with at most five samples the answer here is necessarily a first descriptive read rather than a population-level conclusion. The independent variable is tissue type, tumor versus normal astrocyte-derived brain tissue, and the dependent variable is the distribution of H0 persistence values extracted from each sample's reconstructed three-dimensional chromatin structure.

I hypothesize that:

- *If real H3K27M-driven tumor Hi-C data shows the same bimodal persistence pattern found in simulated data, this will support persistence heterogeneity as a real structural consequence of oncohistone-driven chromatin disruption, not an artifact of the simulation's own assumptions.*
- *If the geometry-only reconstruction method used in the original simulated-data study distorts real, sparser Hi-C data, I will replace it with a method that models physical connectivity between neighboring loci, which should produce measurably different persistence statistics.*

Literature Review

Chromatin architecture and genome folding

The human genome doesn't fold randomly, it's organized into layered three-dimensional structures inside the nucleus. Chromosomes separate into active and inactive compartments. Within these compartments, topologically associating domains form regions where DNA interacts more within a domain than across its boundary, creating functional units of gene regulation [2], [3]. Disrupting these boundaries drives oncogene activation in other glioma subtypes, including CTCF loss in IDH-mutant gliomas [4]. Wang et al. mapped this kind of structural change specifically in H3K27M-driven pediatric high-grade glioma, producing one of the only high-depth, public Hi-C datasets for this tumor type [1]. That dataset, five core samples spanning two H3K27M-mutant tumor lines, one H3-wildtype line, one tumor tissue sample, and one normal astrocyte control, forms the basis of the analysis proposed here.

Persistent homology and reconstruction methods

Persistent homology has been applied to genomic and chromatin data in a small but growing body of work: Sauerwald et al. used it to study chromosome structure during cellular differentiation [6], and more recent multiscale pipelines have called topologically associating domains and loops directly from real Hi-C data [7]. Converting a Hi-C contact matrix into three-

dimensional coordinates is itself unresolved: classical multidimensional scaling treats each pairwise distance independently and ignores the physical constraint that neighboring loci, connected along the same DNA strand, can't be arbitrarily far apart. Li et al. addressed this with a connectivity term between neighboring loci layered onto classical scaling, producing more accurate structures under sparse, noisy conditions than distance-only methods [8]. This work extended H0 persistence to a cancer-versus-normal feature using classical scaling, though only on simulated data; real Hi-C data is far sparser than those earlier simulated matrices, so this proposal adopts the connectivity-aware method in place of classical scaling.

Gaps in Current Literature

Three specific gaps motivate this study. No published research has tested whether persistence-based chromatin heterogeneity, shown only in simulated data, holds up in real tumor tissue, even though external review of my earlier study named exactly this limitation. The geometry-only reconstruction method used in most prior topological work on chromatin hasn't been replaced with a connectivity-aware alternative in this specific disease, and statistical treatment of topological summary statistics under very small sample sizes, an unavoidable limit when working with rare, high-depth pediatric tumor Hi-C data, is rarely addressed directly: most persistent homology work in genomics doesn't report confidence bounds on individual sample estimates at all.

Methodology

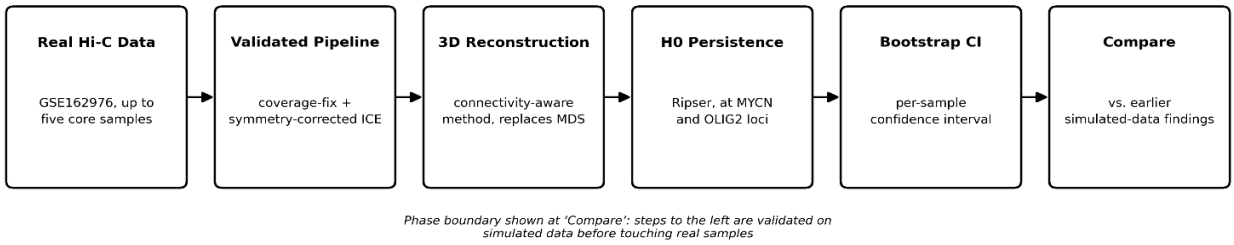


Fig. 1. The proposed six-stage real-data pipeline. The first four stages build on the validated simulated-data pipeline from my earlier study; the connectivity-aware reconstruction step will be implemented and validated against simulated data before being applied to real samples.

Every step below exists to answer one honest question responsibly: given only a handful of real children's tumor samples, can this pipeline say something trustworthy about them, rather than something that only looks precise.

Real Hi-C data

I use Hi-C data from a published pediatric high-grade glioma cohort [1], starting with MYCN and OLIG2, the two loci the source paper itself flags as showing different structure between conditions, with genome-wide analysis planned once the pipeline proves stable. Up to five core samples make up the tumor-versus-normal comparison: two H3K27M-mutant tumor lines, one H3-wildtype line, one tumor tissue sample, and one normal astrocyte control, though the tumor tissue sample's availability hasn't been confirmed yet since raw reads were withheld for patient privacy. All samples were sequenced above 700 million reads, which answers a concern raised

about my earlier study: pipelines built on simulation should calibrate resolution against real coverage, not assume simulated parameters transfer automatically [7]. Given the small sample size, I'll report findings per sample, not pooled, using whichever samples turn out loadable.

Validated pipeline

I'll convert raw contact records into dense contact matrices, using a pipeline I'll build and validate with automated checks and randomized cross-checks against a brute-force reference. A known risk here is low-coverage bins, common in real Hi-C data but absent from simulated data: left unhandled, these can get near-zero contact values that make the distance-transform step blow up mathematically. The pipeline will separate genuinely observed contacts from artificially floored values before computing distance, specifically to avoid this.

3D reconstruction

I'll convert normalized contact matrices into three-dimensional coordinates using the connectivity-aware reconstruction method [8], in place of the classical multidimensional scaling I used in my earlier study. This method isn't yet part of my codebase, so I'll implement it from its published formulation, then validate it against my existing simulated dataset before touching real samples, confirming it reproduces sensible structures where the correct answer is already known, and still finds cancer chromatin more persistent than normal, matching the originally published effect.

H0 persistence

I'll then compute H0 persistent homology using Ripser, following the same steps validated in my earlier study, extracting mean, standard deviation, and total persistence for each sample, the basis for the comparison shown as the final stage in Fig. 1.

Bootstrap confidence intervals

I'll apply bootstrap resampling within each sample, 200 resamples of 80 percent of that sample's genomic bins, following the method Fasy et al. established for building confidence sets on persistence diagrams [9], testing it on synthetic structures before integrating it into the real-data pipeline. This gives each sample its own confidence interval: a real check on reliability given the sparsity of real data, not just a single unchecked number. Within-sample bootstrap variance and between-group biological variance are different things, the first is how much a single patient's number might wobble on resampling, the second is whether patients actually differ. This method measures the first, not the second.

Risk Management and Fallbacks

Four specific technical risks are worth naming, each paired with a fallback if the primary approach fails outright. Low-coverage bins can distort the distance-transform step; if persistence computation still proves unstable at the fine resolution needed for MYCN and OLIG2 despite the coverage-handling fix described above, the fallback is a coarser bin size, standard practice for improving signal density at the cost of detail. The connectivity-aware reconstruction, implemented from a published formulation rather than existing code, carries some risk of subtle errors; if it doesn't validate cleanly against the simulated dataset, or proves unstable on real data anyway, the fallback is classical multidimensional scaling, the same method my earlier study used successfully, which brings back the geometry-only limitation but guarantees a reportable result. The astrocyte control's mismatch with the tumor's cell of origin is addressed below as a

limitation. Sample availability for one tumor line is unconfirmed, so results will reflect whichever subset loads; if bootstrap resampling proves unstable on the sparsest samples, the fallback is reporting point estimates without confidence intervals, marked as reduced rigor.

Potential Limitations

The normal control available in this dataset is normal human astrocytes. H3K27M-driven tumors are now understood to arise mainly from oligodendrocyte precursor cells, not mature astrocytes, though a smaller astrocyte-like tumor subpopulation exists too [11], and I couldn't find a comparably deep, public Hi-C dataset from purified precursor cells. Any observed difference between tumor and normal samples may therefore reflect a mix of cancer-associated disruption and baseline differences between cell types, which I can't fully separate with only this control.

A second limitation is sample size: at most five core samples reach sufficient sequencing depth for this tumor type. Within-sample bootstrap confidence intervals strengthen each individual estimate, but they don't replace a population-level significance test, so I report results as descriptive and hypothesis-generating for this reason, not as a confirmed population comparison. This study doesn't extend to higher-dimensional persistence (H1, H2), uninterpretable at this resolution.

Project Practicalities

- *Phase 1 (Weeks 1-3): retrieve the core Hi-C files, confirm data integrity, then implement the connectivity-aware reconstruction method and build the bootstrap confidence-interval procedure, validating both against my existing simulated dataset before touching real data.*
- *Phase 2 (Weeks 4-6): run the completed pipeline on real data with the coverage-handling fix, reconstruct three-dimensional structures using the connectivity-aware method, and compute H0 persistence with bootstrap confidence intervals at the MYCN and OLIG2 loci for each confirmed sample.*
- *Phase 3 (Weeks 7-9): compare per-sample results against the simulated-data findings, document where they agree or diverge, and extend analysis genome-wide for samples where the priority-loci results support doing so.*
- *Phase 4 (Weeks 10-12): finalize the written report.*

Post-Program Plan

I'll document the validated pipeline for reuse on other pediatric brain tumor Hi-C cohorts as more data becomes available, then, if the signature holds in real tissue, prepare it for peer-reviewed submission, testing cross-tumor-location generalization and prognostic value along the way, and eventually whether persistence heterogeneity tracks with survival.

Conclusion

The earlier phase of this work found an unpredicted bimodal pattern in cancer chromatin. This proposal is the first honest test of whether that pattern survives contact with real tissue, not a statistically powered confirmation. If it holds, this is among the first biomarker candidates here; if not, that is equally informative for deciding whether persistence-based methods are worth pursuing further in real chromatin data.

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